

ST JOHNS WEST, NL, Canada

WMO: 712500

Lat: 47.5133N

Lon: 52.7833W

Elev: 361

StdP: 14.51

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.0	11.4	0.0	5.6	11.6	3.3	6.6	14.4	31.9	29.0	28.9	29.8	9.4	290	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	78.7	68.0	76.0	66.4	73.4	64.9	71.0	75.5	69.1	73.2	67.3	71.3	11.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	109.3	73.5	67.6	102.8	71.8	65.6	95.9	69.9	35.1	75.7	33.5	73.3	32.0	71.3	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.3	19.9	DB	2.8	83.8	3.6	3.1	0.3	86.1	-1.9	87.9	-3.9	89.7	-6.5	91.9	(4)
(5)			WB	2.2	73.7	3.5	2.3	-0.3	75.3	-2.4	76.6	-4.4	77.9	-6.9	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	26.8	25.6	28.9	36.7	44.3	52.4	62.7	62.9	56.2	47.1	39.5	31.3	(6)
(7)		DBStd	14.64	7.53	7.48	6.76	6.29	6.56	7.45	6.56	5.45	6.36	6.80	7.00	7.60	(7)
(8)		HDD50	3779	719	682	654	400	205	58	3	0	15	141	322	580	(8)
(9)		HDD65	8141	1183	1102	1119	848	642	384	122	105	272	556	764	1044	(9)
(10)		CDD50	1214	0	0	0	3	28	129	396	399	200	50	8	1	(10)
(11)		CDD65	100	0	0	0	0	0	5	50	38	7	0	0	0	(11)
(12)		CDH74	472	0	0	0	0	2	33	258	147	32	0	0	0	(12)
(13)		CDH80	43	0	0	0	0	0	1	27	10	5	0	0	0	(13)
(14)	Wind	WSAvg	9.6	10.7	11.5	11.1	10.0	9.1	8.5	8.2	7.9	8.5	9.3	9.9	10.8	(14)
(15)	Precipitation	PrecAvg	66.1	7.6	6.7	6.3	5.2	4.4	3.8	3.1	4.5	4.7	6.1	6.3	6.8	(15)
(16)		PrecMax	78.7	12.0	13.1	12.6	9.4	8.0	7.2	5.9	12.1	8.6	11.0	11.0	12.0	(16)
(17)		PrecMin	54.9	4.2	2.3	3.0	1.0	0.9	0.9	0.8	1.5	1.0	2.8	3.4	3.7	(17)
(18)		PrecStd	6.7	2.2	2.6	2.2	2.4	1.9	1.5	1.4	2.5	1.9	2.2	2.0	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	49.3	50.8	61.8	70.2	77.6	83.1	81.2	78.0	68.9	62.4	56.0	(19)
(20)		0.4%	MCWB	52.3	48.1	47.1	53.6	59.4	66.7	70.7	68.1	68.0	65.0	60.1	54.2	(20)
(21)		2%	DB	46.3	43.9	45.6	55.9	65.2	73.7	79.8	77.7	72.9	64.8	58.7	51.0	(21)
(22)		2%	MCWB	45.4	43.2	42.5	50.4	56.5	63.7	68.8	67.5	66.0	61.6	57.2	50.0	(22)
(23)		5%	DB	40.7	39.4	41.0	51.1	60.7	69.7	76.9	75.2	69.6	61.6	55.0	45.9	(23)
(24)		5%	MCWB	39.8	38.2	38.4	46.4	53.1	60.9	67.3	66.2	64.0	58.6	53.6	44.8	(24)
(25)		10%	DB	36.8	35.9	37.6	46.8	56.2	65.7	74.0	72.6	66.7	58.0	51.0	41.6	(25)
(26)		10%	MCWB	35.8	34.6	35.5	43.0	49.8	59.0	64.9	65.0	61.8	55.0	49.5	40.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	48.3	48.4	57.6	63.1	70.2	74.4	72.7	70.9	66.2	60.7	55.4	(27)
(28)		0.4%	MCD B	53.4	49.2	50.4	60.0	67.3	75.3	80.6	77.8	74.6	68.2	61.8	56.1	(28)
(29)		2%	WB	45.3	43.3	43.2	51.5	58.8	66.9	71.7	70.3	67.8	62.6	57.2	50.1	(29)
(30)		2%	MCD B	46.1	43.9	45.1	54.3	63.3	71.0	76.8	74.3	71.2	63.8	58.4	51.1	(30)
(31)		5%	WB	40.1	38.5	39.0	47.2	54.4	63.2	69.8	68.4	65.2	58.9	53.8	45.0	(31)
(32)		5%	MCD B	40.7	39.2	40.5	50.1	58.3	67.5	74.1	72.2	68.7	60.9	54.9	45.9	(32)
(33)		10%	WB	36.0	34.8	35.8	43.4	50.7	59.6	67.7	66.5	62.7	55.6	49.7	40.5	(33)
(34)		10%	MCD B	36.7	35.7	37.3	46.4	55.3	64.8	71.8	70.9	65.9	57.7	50.8	41.3	(34)
(35)	Mean Daily Temperature Range		MDBR	12.1	12.4	11.9	13.1	15.5	16.0	16.5	15.1	12.4	11.7	10.9	(35)	
(36)		5% DB	MCD BR	18.0	17.1	16.6	20.4	23.3	22.2	19.7	18.6	17.8	16.0	17.1	16.8	(36)
(37)			MCWBR	18.4	17.3	14.5	15.5	15.3	13.7	11.2	11.0	11.9	13.4	16.5	16.8	(37)
(38)			MCD BR	18.1	16.9	16.0	18.7	20.2	19.5	16.7	15.9	16.4	15.3	16.9	17.0	(38)
(39)		5% WB	MCWBR	18.6	17.4	15.2	15.7	15.5	13.7	11.2	11.4	12.5	13.9	16.5	17.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.318	0.336	0.367	0.371	0.403	0.433	0.399	0.371	0.346	0.325	0.313	(40)	
(41)		taud	2.354	2.423	2.413	2.365	2.414	2.347	2.303	2.413	2.461	2.487	2.457	2.371	(41)	
(42)		Ebn at Noon	239	269	281	281	283	273	263	268	266	255	237	225	(42)	
(43)		Edh at Noon	23	26	31	36	36	38	40	34	30	25	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	299	545	917	1246	1528	1613	1621	1369	1091	663	375	250	(44)	
(45)		RadStd	22	41	71	124	125	126	116	87	68	60	50	21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	-1.28	N/A	N/A	N/A	N/A	N/A	N/A	+495	-230	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page